SOLVING MATRIX EQUATIONS

LINEAR ALGEBRA

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Interpretations of Matrix Multiplication

Solving Vector Equations

Solving Matrix Equations

INTERPRETATIONS OF MATRIX MULTIPLICATIONS

INTERPRETATION 1

$$\begin{array}{ccc} A & B & C \\ \begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix} \cdot & \begin{bmatrix} 1 & -2 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} = & \begin{bmatrix} 11 & 12 \\ 15 & 23 \end{bmatrix} \end{array}$$

The $i\,j$ entry of C is the **dot product** of row i in A and column j in B

$$c_{ij} = \overrightarrow{a_i} \cdot \overrightarrow{b_j}$$

$$c_{11} = \overrightarrow{a_1} \cdot \overrightarrow{b_1}$$

$$c_{11} = (2 \cdot 1) + (3 \cdot 2) + (1 \cdot 3) = 11$$

$$c_{21} = \overrightarrow{a_2} \cdot \overrightarrow{b_1}$$

$$c_{21} = (-4 \cdot 1) + (2 \cdot 2) + (5 \cdot 3) = 15$$

INTERPRETATION 1

$$\begin{array}{ccccc} A & & B & & C \\ \begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix} & \cdot & \begin{bmatrix} 1 & -2 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} & = & \begin{bmatrix} 11 & 12 \\ 15 & 23 \end{bmatrix} \end{array}$$

The $i\,j$ entry of C is the **dot product** of row i in A and column j in B

$$c_{ij} = \overrightarrow{a_i} \cdot \overrightarrow{b_j}$$

$$c_{12} = \overrightarrow{a_1} \cdot \overrightarrow{b_2}$$

$$c_{11} = (2 \cdot -2) + (3 \cdot 5) + (1 \cdot 1) = 12$$

$$c_{22} = \overrightarrow{a_2} \cdot \overrightarrow{b_2}$$

$$c_{22} = (-4 \cdot -2) + (2 \cdot 5) + (5 \cdot 1) = 23$$

INTERPRETATION 2

$$\begin{array}{ccccc} A & & B & & C \\ \begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix} & \cdot & \begin{bmatrix} 1 & -2 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} & = & \begin{bmatrix} 11 & 12 \\ 15 & 23 \end{bmatrix} \end{array}$$

The j-th column of C is the product of **A** and the j-th **column** of **B**

$$\overrightarrow{c_j} = A \cdot \overrightarrow{b_j}$$

$$\begin{array}{ccccc} A & & \overrightarrow{b_1} & & \vec{c}_1 \\ \begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix} & \cdot & \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} & = & \begin{bmatrix} 11 \\ 15 \end{bmatrix} \end{array}$$

INTERPRETATION 2

$$\begin{array}{ccccc} A & & B & & C \\ \begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix} & \cdot & \begin{bmatrix} 1 & -2 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} & = & \begin{bmatrix} 11 & 12 \\ 15 & 23 \end{bmatrix} \end{array}$$

The j-th column of C is the product of A and the j-th **column** of **B**

$$\overrightarrow{c_j} = A \cdot \overrightarrow{b_j}$$

$$\begin{array}{ccccc} A & & \overrightarrow{b_2} & & \vec{c}_2 \\ \begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix} & \cdot & \begin{bmatrix} -2 \\ 5 \\ 1 \end{bmatrix} & = & \begin{bmatrix} 12 \\ 23 \end{bmatrix} \end{array}$$

EXERCISE

Given that $\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 13 \\ 19 \\ 25 \end{bmatrix}$,

and $\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \\ 17 \end{bmatrix}$,

and $\begin{bmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{bmatrix}\begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 27 \end{bmatrix}$

find $\begin{vmatrix} 4 & 5 \\ 6 & 7 \\ 8 & 9 \end{vmatrix}\begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \end{vmatrix}$.

The sum of scalar multiples of vectors

$$\boldsymbol{k_1\overrightarrow{v_1} + k_2\overrightarrow{v_2} + \cdots k_n\overrightarrow{v_n}}$$

$\boldsymbol{k}$ = coefficient

$$\boldsymbol{3}\begin{bmatrix}\mathbf{1}\\\mathbf{2}\\\mathbf{3}\end{bmatrix} + \boldsymbol{4}\begin{bmatrix}\mathbf{-2}\\\mathbf{5}\\\mathbf{0}\end{bmatrix} \boldsymbol{-2}\begin{bmatrix}\mathbf{4}\\\mathbf{1}\\\mathbf{-1}\end{bmatrix} = \begin{bmatrix}\mathbf{-13}\\\mathbf{24}\\\mathbf{11}\end{bmatrix}$$

INTERPRETATION 3

The $j\text{-}th$ column of C is the **linear combination** of the columns of A with the coefficients given by the entries in the $j\text{-}th$ column of B

$$\overrightarrow{c_j} = b_{1j}\overrightarrow{a_1} + b_{2j}\overrightarrow{a_2} + \cdots b_{nj}\overrightarrow{a_n}$$

$$\underset{A}{\begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix}} \cdot \underset{B}{\begin{bmatrix} 1 & -2 \\ 2 & 5 \\ 3 & 1 \end{bmatrix}} = \underset{C}{\begin{bmatrix} 11 & 12 \\ 15 & 23 \end{bmatrix}}$$

$$1\begin{bmatrix} 2 \\ -4 \end{bmatrix} + 2\begin{bmatrix} 3 \\ 2 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ 15 \end{bmatrix}$$

INTERPRETATION 3

The j-th column of C is the **linear combination** of the columns of A with the coefficients given by the entries in the j-th column of B

$$\overrightarrow{c_j} = b_{1j}\overrightarrow{a_1} + b_{2j}\overrightarrow{a_2} + \cdots b_{nj}\overrightarrow{a_n}$$

$$\begin{array}{ccc} A & B & C \\ \begin{bmatrix} 2 & 3 & 1 \\ -4 & 2 & 5 \end{bmatrix} \cdot & \begin{bmatrix} 1 & -2 \\ 2 & 5 \\ 3 & 1 \end{bmatrix} = & \begin{bmatrix} 11 & 12 \\ 15 & 23 \end{bmatrix} \end{array}$$

$$-2\begin{bmatrix} 2 \\ -4 \end{bmatrix} + 5\begin{bmatrix} 3 \\ 2 \end{bmatrix} + 1\begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 23 \end{bmatrix}$$

For $\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{2} \\ \mathbf{6} & \mathbf{-5} \\ \mathbf{3} & \mathbf{4} \end{bmatrix}$, write down a **matrix B** such that

$$\boldsymbol{AB} = [2\overrightarrow{a_1} + 3\overrightarrow{a_2} \quad -3\overrightarrow{a_1} + 4\overrightarrow{a_2} \quad \overrightarrow{a_1} + \overrightarrow{a_2} \quad 5\overrightarrow{a_2}]$$

SOLVING VECTOR EQUATIONS

Solving vector equation of the form

$$\boldsymbol{A\vec{x} = \vec{b}}$$

where :

A = matrix

$\vec{x}$ = unknown vector

$\vec{b}$ = known vector

Solve the equation $\boldsymbol{A\vec{x} = \vec{b}}$ for $\vec{x}$, where

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{-2} \\ \mathbf{3} & \mathbf{4} & \mathbf{-5} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} \end{bmatrix} \text{ and } \vec{\boldsymbol{b}} = \begin{bmatrix} \mathbf{1} \\ \mathbf{7} \\ \mathbf{-2} \end{bmatrix}$$

SOLVING MATRIX EQUATIONS

Solving matrix equation of the form

$$\boldsymbol{AX} = \boldsymbol{B}$$

where :

A = square matrix

X = unknown matrix

B = known matrix

Solve the matrix equation $\boldsymbol{AX} = \boldsymbol{B}$ for **matrix X** , where

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{3} \\ \mathbf{2} & \mathbf{2} \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} \mathbf{1} & \mathbf{4} & \mathbf{1} \\ \mathbf{-1} & \mathbf{2} & \mathbf{1} \end{bmatrix}.$$

SEATWORK